

BU LearnSpace

A Digital Academic Platform for Bangladesh University

A Web-Based Academic Management System for
Admin, Teacher and Student Portals

Technology Stack: PHP · MySQL · HTML/CSS · PHPMailer
Prepared for Project Proposal & Presentation Use

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1. Executive Summary

BU LearnSpace is a web-based academic management platform designed for Bangladesh University to digitize day-to-day academic operations for administrators, teachers, and students. The system replaces manual, paper-based workflows for study material distribution, assignment submission, grading, attendance tracking, and academic record management with a single, centralized, role-based web application.

The platform is built using PHP and MySQL following a role-based architecture with three independent portals — Admin, Teacher, and Student — each offering a tailored dashboard and feature set relevant to that user's responsibilities. It provides secure authentication, file management, automated academic record-keeping, and email-based password recovery.

Aspect	Details
Project Name	BU LearnSpace
Tagline	A Digital Academic Platform for Bangladesh University
Platform Type	Web-based Academic Management System (LMS-style)
User Roles	Admin, Teacher, Student
Core Technology	PHP, MySQL (MySQLi), HTML5, CSS3, PHPMailer
Architecture	Role-based multi-portal web application with session authentication
Deployment Target	Local/Campus server via XAMPP-compatible PHP + MySQL stack

2. Problem Statement & Motivation

Universities that rely on manual or fragmented processes for sharing study materials, collecting assignments, recording marks, and tracking attendance face several recurring challenges:

- Study materials are shared informally (email, messaging apps), making them hard to organize, search, or track.
- Assignment collection and grading is manual, slow, and prone to lost submissions.
- Attendance is recorded on paper registers, making reporting and record-keeping difficult.
- Marks and assessment records are scattered across spreadsheets with no student-facing visibility.
- There is no single, secure, role-based system for admins to manage users and departments centrally.

BU LearnSpace addresses these problems by providing one unified, web-accessible platform where each stakeholder — administrator, teacher, and student — has a dedicated, permission-controlled workspace for these exact tasks.

3. Project Objectives

- Provide a centralized, digital repository for course study materials, organized by department, course, and semester.
- Enable teachers to create, distribute, and grade assignments online, with deadline tracking.
- Allow teachers to record class-wise attendance and let students view their own attendance history.
- Provide a structured assessment and marks management system visible to both teachers and students.
- Give administrators full control over user accounts, roles, and department structures.
- Ensure secure, role-based authentication with encrypted password storage and email-based recovery.

4. System Users & Roles

Role	Responsibilities / Access
Administrator	Manage teacher and student accounts (create, edit, delete), assign departments and batches, and oversee the overall platform through a dedicated admin dashboard.
Teacher	Upload and manage study materials, create assignments and review/grade student submissions, create assessments and enter marks, record attendance, and manage a personal profile.
Student	Browse and download study materials, bookmark and rate materials, submit assignments, view grades/marks and feedback, view personal attendance records, and manage a personal profile.

5. Core Modules & Features

5.1 Admin Module

- Secure admin login separate from teacher/student authentication.
- Centralized dashboard to search, add, edit, and delete user accounts (teachers and students).
- Assign or update a user's department and student batch information.
- Department-based organization of the entire student and teacher base.

5.2 Study Materials Module

- Teachers upload course materials (PDF/image files) with title, course name, semester, department, and an optional cover image.
- Students can browse, preview, and download materials filtered by department, course, and semester.
- Automatic view-count and download-count tracking per material for popularity insight.
- Student-side rating (1–5 stars) and bookmarking system for quick access to saved materials.
- Teachers can edit or delete their own uploaded materials at any time.

5.3 Assignment Module

- Teachers create assignments with title, description, course, semester, department, and a submission deadline.
- Students upload their assignment files directly through the student portal before the deadline.
- Teachers review submissions, assign marks, and provide written feedback per student.
- Dashboard indicator for "pending reviews" helps teachers track ungraded submissions.

5.4 Assessment & Marks Module

- Teachers create assessments (quizzes, tests) per course/semester/department with a configurable total mark.
- Marks are recorded per student with optional remarks/feedback.
- Students can view their obtained marks and remarks for every assessment in their courses.

5.5 Attendance Module

- Teachers create a dated attendance session for a course/semester/department.
- Attendance is recorded per student as Present, Absent, or Late.
- Students can view their own attendance history and track class-wise participation.

5.6 Account & Security Features

- Passwords are stored using industry-standard bcrypt hashing (never in plain text).
- Role-based session authentication keeps Admin, Teacher, and Student areas fully isolated.
- "Forgot Password" self-service flow for teachers and students, powered by PHPMailer for secure email delivery of reset instructions.
- Editable user profiles including profile photo upload for teachers and students.

6. Technology Stack

Layer	Technology Used
Frontend	HTML5, CSS3 (custom stylesheet), responsive card/dashboard-based UI
Backend	PHP (server-side scripting, procedural with mysqli prepared statements)
Database	MySQL — relational schema with 11 core tables and foreign-key relationships
Authentication	PHP Sessions + bcrypt password hashing
Email Service	PHPMailer (Composer package) for password-reset emails
Dependency Manager	Composer (composer.json / composer.lock)
File Storage	Server-side file system (/uploads) for materials, submissions, and profile photos
Version Control	Git
Local Dev Environment	XAMPP-compatible stack (Apache + MySQL + PHP)

7. Database Design Overview

The system uses a normalized relational schema comprising 11 interconnected tables that model users, departments, academic content, and academic records:

Table	Purpose
users	Stores all Admin, Teacher, and Student accounts with role, department, and hashed password.
departments	Master list of academic departments used across the platform.
materials	Study material records: file, cover image, course, semester, view/download counters.
material_ratings	Student star ratings (1–5) for individual study materials.
material_bookmarks	Student bookmarks/saved materials for quick access.
assignments	Assignment definitions created by teachers, with deadlines.
submissions	Student assignment submissions, with marks and feedback.
assessments	Quiz/test definitions with total marks, per course/semester.
assessment_marks	Student marks and remarks for each assessment.
attendance	Dated attendance sessions per course/semester/department.
attendance_records	Per-student attendance status (present/absent/late) for a session.

All tables enforce referential integrity through foreign keys (e.g., materials → departments, submissions → assignments/users), and cascading deletes are used where appropriate (e.g., deleting an assignment removes its submissions) to keep data consistent.

8. System Architecture

BU LearnSpace follows a classic three-tier web architecture:

Tier	Description
Presentation Layer	HTML/CSS pages rendered per role (Admin, Teacher, Student), served dynamically by PHP.
Application Layer	PHP scripts handle authentication, business logic, form processing, file uploads, and email dispatch (PHPMailer).
Data Layer	MySQL relational database storing users, academic content, and records; accessed via mysqli prepared statements.

Each of the three portals (index.php entry point → admin/, teacher/, student/ directories) is independently authenticated, so a session for one role cannot access another role's pages, enforcing strict access control across the application.

9. Key Highlights

Feature	Benefit
3 dedicated portals	Clear separation of responsibilities and permissions per user type.
Centralized materials hub	One searchable place for all course content, replacing scattered file sharing.
Online assignment workflow	End-to-end digital submission and grading cycle with feedback.
Digital attendance	Replaces paper registers; instantly viewable by students.
Transparent marks system	Students see assessment results and remarks in real time.
Secure by design	Bcrypt password hashing, session isolation, prepared SQL statements.

10. Future Enhancement Opportunities

- Real-time notifications for new materials, assignments, and grading updates.
- Analytics dashboards showing attendance trends and academic performance over time.
- Online quiz/exam module with auto-grading for objective questions.
- Mobile-responsive or dedicated mobile app version of the portals.
- Role for department heads/coordinators with cross-teacher reporting.
- Integration with cloud storage for scalable file handling.

11. Conclusion

BU LearnSpace consolidates the core academic workflows of a university — content sharing, assignments, assessments, and attendance — into a single, secure, role-based web platform. By digitizing these processes, the system reduces administrative overhead, improves transparency between teachers and students, and creates a scalable foundation that can grow with the institution's future academic technology needs.